

Homework Grading Report

Student:	Student_1
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: colSums(), na.omit()
Part 3: Basic Data Transformation	15	15.0	
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	All tasks completed
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	119.3	95%

Instructor Assessment

You have demonstrated that you can import, clean, reshape, and join multiple datasets using tidyverse functions, which is a solid technical foundation. However, your reflection responses are overly brief and do not connect the analytical steps to concrete business insights, limiting the value of your work for decision-makers. Focus on embedding specific results and their implications into your written explanations, and consider adding visual summaries to make your findings more actionable.

What You Did Well

Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.2: Reshape to Wide Format Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.3: Reshape Back to Long Format Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.1: Clean Text

Issues

Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.1: Check for Missing Values - Your code attempts to create 'missing_summary' but fails because 'sales_data' is not defined. You need to load the dataset first before running this task. The correct approach would be to ensure 'sales_data' exists in your environment before attempting to analyze it.

Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.2: Parse Dates and Extract Components - Your code produces a warning indicating that all formats failed to parse for the date column. This suggests that the date format in 'Sale_Date' doesn't match what 'mdy()' expects. You should check the actual format of the dates in your data and adjust accordingly, possibly using 'ymd()' or another appropriate parsing function depending on how the dates are formatted.

To strengthen your data cleaning, you need to quantify the impact of removed rows on key metrics; for example, calculate total revenue before and after cleaning and report the percentage change. This matters because stakeholders need to know whether cleaning materially alters business results. Add a line such as `revenue_before <- sum(sales_data$Revenue, na.rm = TRUE)` and compare it to `revenue_after <- sum(sales_data_cleaned$Revenue)`; the solution shows a 2% drop after outlier removal, which informs confidence in the cleaned dataset.

Your regional and category summaries would be more powerful if you included visualisations (e.g., bar charts) or highlighted the top-3 performers with explicit values. Visual cues help executives quickly grasp where the biggest opportunities lie. Use `ggplot2` to plot `region_summary` and annotate the region with the highest `total_revenue`.

When reshaping data, you should demonstrate why you chose long vs. wide by performing a downstream analysis that requires each format. For instance, after `pivot_wider` you could compute a correlation matrix across product categories, while the long format could feed a time-series plot. Adding a short comment before each reshape explaining the intended analysis would make the purpose clear.

Your explanation of join types lacks reference to data integrity checks; you should verify that the left join did not introduce duplicate rows and that the inner join did not unintentionally drop needed records. This matters because join errors can distort KPI calculations. Include a step like `anti_join(customers, orders, by = "CustomerID")` to identify customers without orders, and discuss the findings.

Reflection answers need depth: move beyond stating that cleaning improves accuracy to describing specific changes observed in your dataset. This matters because it demonstrates critical thinking and the ability to translate technical work into business impact. Revise each answer to include at least one concrete example from your analysis, such as "removing 12 outlier transactions reduced total revenue by \$45,000, which shifted the top-selling region from West to North."

Reflection & Critical Thinking

You answered all 5 reflection questions, but each response is only one sentence and largely repeats textbook definitions without linking to your own analysis. For Question 9.1 you note that cleaning improves accuracy, yet you do not describe how the removal of NAs or outliers changed any of your summary metrics. To deepen your answer, reference specific numbers (e.g., rows removed, changes in

total revenue) and explain the business impact of those changes. In Question 9.2 you correctly state that grouped summaries reveal trends, but you do not cite any concrete insight such as which region had the highest average revenue or which product category drove the most sales. Including those specific findings would demonstrate that you can translate raw numbers into actionable business recommendations. Your answer to Question 9.3 identifies the general purpose of wide vs. long formats, yet you miss an opportunity to tie the formats to the actual tasks you performed (e.g., using wide format for a side-by-side regional performance dashboard). Adding a concrete scenario from your own work would show you understand when each shape is most useful. For Question 9.4 you correctly differentiate `left_join` and `inner_join`, but you do not explain why you chose a left join for customers-orders and an inner join for orders-order_items in your code. Describing the business rationale-preserving all customers even if they have no orders, versus focusing only on orders that have line items-would strengthen your response. In Question 9.5 you argue that joining datasets is the most valuable skill, which is reasonable, yet you do not connect this claim to a specific insight you derived (e.g., linking customer demographics to purchase behavior). Providing a concrete example from your analysis would make your argument more persuasive.

Business Application

The regional_summary and category_summary tables you created provide a clear view of where revenue is concentrated, which is exactly the type of insight senior management needs for resource allocation. Highlighting the top-performing region or product category would allow the company to focus marketing spend where it yields the highest return. Your performance_tier categorisation (High, Medium, Low) translates raw revenue numbers into a business-friendly segmentation that can be used for targeted promotions or sales incentives. This step bridges the gap between technical analysis and actionable strategy.

Recommendations

Complete a short post-analysis audit where you compare key metrics before and after each transformation step; document any shifts and explain their business relevance. Practice writing narrative summaries that tie each code block to a business question, using specific numbers from your outputs to support your conclusions.

Technical Summary

- Completion: 8 out of 12 sections (67%). Calculated score: 67%.
- Completed: Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values, Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue, Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue, Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations, Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.2: Reshape to Wide Format, Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.3: Reshape Back to Long Format, Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.1: Clean Text Data, Part 8: Advanced Wrangling & Business Intelligence (Lesson 8) - Task 8.2: Calculate Business KPIs
- Incomplete: Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.1: Check for Missing Values, Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.2: Parse

Dates and Extract Components

- Completion: 8/9 sections (89%). Score: 95%
- Variables found: 22/22
- Output accuracy: 95.7% (22/23 checks passed)

Report generated February 08, 2026 at 08:28 PM by AI Homework Grading System